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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations
December 2025

ME2002-1 – INDUSTRIAL ROBOTICS AND AUTONOMOUS SYSTEMS
(For MEC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Enlist the main elements of a hydraulic system used in robot and explain their functions briefly.
A) Pipes, wires, controllers. B) Pump, reservoir, valves, actuators, and hoses.
C) Pump, reservoirs D) Sensors, motors, batteries.
2. Explain the performance and selection criteria of electric motors in robotics
A) Torque, speed, accuracy, load capacity. B) Weight and color.
C) Fuel and heat capacity. D) Cost only.
3. Define a robot.
A) A reprogrammable, multifunctional manipulator used for various tasks. B) A machine that can only perform a single pre-programmed task.
C) Any mechanical device controlled by a remote. D) A manually operated device for lifting objects.
4. Which type of gripper uses the force of compressed air?
A) Electric gripper B) Hydraulic gripper
C) Magnetic gripper D) Pneumatic gripper
5. What is the main function of a robot gripper?
A) Measure environmental conditions B) Connect the robot to its power supply
C) Hold, grip, or manipulate objects D) Generate the control program
6. What is DH convention.
A) A systematic method to assign coordinate frames to robot links and joints. B) A manual way to control robot arms without coordinates.
C) A programming language used in robotics. D) A method for drawing freehand sketches of robots.
7. Which of the following is not a standard Denavit-Hartenberg (DH) parameter?
A) Link length (a) B) Link twist (α)
C) Joint offset (d) D) Joint torque (τ)
8. In DH parameters, the 'a' parameter denotes:
A) Distance between two joint axes measured along the common normal B) Angle between two joint axes
C) Offset along the previous z-axis D) Joint angle
9. Redundancy in autonomous systems is used to
A) Reduce system cost. B) Ensure reliability and fault tolerance.
C) Minimize storage requirements. D) Eliminate the need for sensors.

10. Which of the following is a fail-safe design principle?
 - A) The system continues to operate safely even after a failure.
 - B) The system shuts down permanently after minor errors.
 - C) The system ignores faults completely.
 - D) The system operates only when failures occur.
11. Which of the following represents the earliest form of cloud computing?
 - A) Grid computing.
 - B) Quantum computing.
 - C) Blockchain.
 - D) Manual computing.
12. Which of the following is NOT a cloud service model?
 - A) IaaS.
 - B) SaaS.
 - C) PaaS.
 - D) BIOS.
13. Which cloud service provides software applications over the internet?
 - A) IaaS.
 - B) SaaS.
 - C) PaaS.
 - D) VPN.
14. Which sensor combination best demonstrates data fusion?
 - A) GPS and IMU work together for navigation.
 - B) Only one ultrasonic sensor.
 - C) Only one temperature sensor.
 - D) A manual steering wheel.
15. In Cartesian space trajectory planning, motion is planned in terms of
 - A) Joint angles only
 - B) Gripper force
 - C) Position and orientation in 3D space
 - D) Robot weight
16. Which trajectory planning method ensures continuous position, velocity, and acceleration?
 - A) Non-linear interpolation
 - B) Cubic polynomial interpolation
 - C) Constant velocity motion
 - D) Stop-and-go motion Joint space
17. Which among these devices work on CRT principle?
 - A) Black and white Videocon camera.
 - B) Charge coupled devices.
 - C) Solid-state camera.
 - D) Charge injection devices.
18. Which of the following is not typically part of a machine vision system?
 - A) Illumination
 - B) Gripper
 - C) camera
 - D) Image processing software
19. In a machine vision system, frame grabbers are used to:
 - A) Hold the robot arm in place
 - B) Physically grab objects
 - C) Digitize and transfer video frames from the camera to the computer
 - D) Adjust camera focus automatically
20. What is the primary function of a machine vision system?
 - A) To control the robot's gripper force
 - B) To interpret and analyze visual data for decision-making
 - C) To generate power for robotic arms
 - D) To clean the camera lenses automatically

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define industrial robotics and Industrial automation? What are the types of industrial automation types?	6	L*1	1	1
	b) A DC servomotor used to drive a robot joint has a torque constant, 1.25 N-m/A and the voltage constant of 12×10^{-3} V/rpm. The armature resistance of 2.5 Ω . A voltage of 25 V is applied at a point of time of robot cycle when the joint is stationary. Determine, a) The torque of the motor immediately after the voltage is applied. b) The back e.m.f and the respective torque corresponding to rotational speeds of 250 rpm and 500 rpm.	6	L3	2	2

- c) List and explain various types of gripper mechanism 4 L1 1 1
2. a) Briefly Explain different types of actuators used in robot joints 6 L1 1 1
 b) With a neat sketch explain link & joint parameters 6 L1 2 1
 c) For the coordinate systems shown in Fig. 3Q (b) determine the homogeneous transformation matrices 0T_A and 0T_B

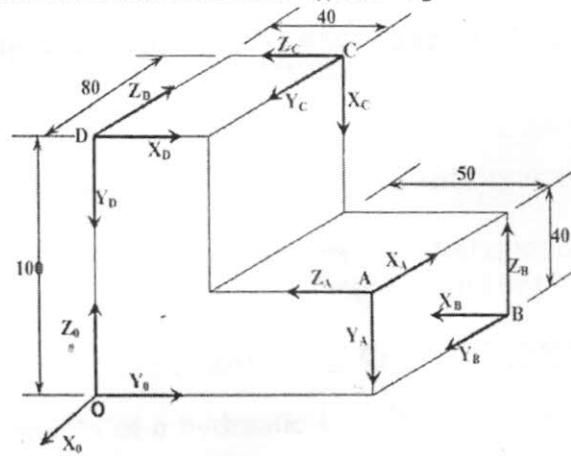


Fig. 2Q (c)

4 L3 2 2

3. a) With the help of neat sketches explain the various configurations used in a robot. 8 L1 1 1
 b) Derive the operation of rotation matrix about an arbitrary axis which is represented by a vector and gives its geometric interpretation. 8 L1 2 1

Unit – II

4. a) Explain trajectory planning technique which uses a cubic polynomial trajectory for a robot joint, for point-to-point applications without via points. 8 L1 4 1
 b) What are the key cloud service requirements to ensure seamless operation, security, and efficiency in AI-driven autonomous systems? 8 L1 3 1
5. a) The motion of a joint of a 3 DOF arm is constrained by an actuator that can produce maximum acceleration/deceleration of 0.35 rad/s^2 and maximum velocity of 15 rad/s . If the trapezoidal velocity profile is assumed for this joint, plot the joint trajectory if joint moves from 0° to 180° in 10 seconds. 8 L2 4 2
 b) What redundancy techniques are commonly applied to improve the reliability and fail-safe performance of autonomous systems? 8 L1 3 1
6. a) Define the following terminology with respect to trajectory planning
 1. Path
 2. Trajectory
 3. Spline
 4. Knott points 8 L1 4 1
 b) How can data fusion and computer vision be integrated to enhance the perception accuracy of autonomous systems? 8 L1 3 1

Unit – III

7. a) Explain robot work cell control and interlocks with an example. 8 L1 4 1
 b) Explain different segmentation techniques used in image processing for robotic applications. 8 L1 5 1

ME2002-1

SEE – December 2025

8. a) With reference to robot programming, explain the different categories of commands used for controlling robot motion, speed, position, and effectors, and sensors. Illustrate your answer with suitable statements (e.g., APPRO, MOVE, SPEED, HERE, OPEN, SIGNAL, etc.) and briefly describe their purpose.
- b) Describe the principal functions of robot vision system.

8	L3	5	1
8	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
